
Addressing the Cold-Start Problem in Recommendation Systems with Explainable Multi-Modal Large Language Models

Balaj Khalid
bkkhalid@usc.edu

Athena Lobo
amlobo@usc.edu

Abstract

Recommendation systems play a critical role in helping users discover relevant products, services, and local businesses. A long-standing challenge in these systems is the cold-start problem, where limited or nonexistent user-item interactions prevent traditional collaborative filtering and content-based models from learning reliable preferences. Recent advances in multimodal large language models (LLMs) create new opportunities to address this challenge by integrating diverse content sources such as textual descriptions, reviews, images, and structured metadata. Unlike earlier multimodal deep learning approaches, LLMs can not only fuse heterogeneous signals but also generate natural-language rationales that improve transparency and user trust. However, existing multimodal LLM-based recommenders often suffer from ineffective modality fusion and weak grounding, leading to noisy representations and reduced performance. In this work, we propose an explainable multimodal LLM framework that combines BERT-based text encoders, CLIP visual features, structured metadata embeddings, and deep cross-modal fusion layers to generate unified item representations tailored for cold-start scenarios. The fused embeddings are integrated into a parameter-efficiently tuned Llama-based reasoning module to produce both ranking predictions and human-readable explanations. Using the Amazon Kindle Store dataset, we demonstrate that our approach improves Precision, Recall, and NDCG compared to strong hybrid and generative cold-start baselines, highlighting the potential of grounded multimodal LLMs for real-world recommendation environments.

1 Introduction

Modern digital platforms rely heavily on recommendation systems to help users navigate overwhelming volumes of online content and products. As marketplaces such as Amazon continue to grow, with millions of new users and items appearing each month, the ability to generate relevant and trustworthy recommendations becomes essential for user engagement and platform profitability. Classical recommender systems typically fall into two broad paradigms: content-based filtering (CB) and collaborative filtering (CF) [7]. Content-based methods learn a user profile from item features and past preferences, recommending items with similar attributes [12]. Collaborative filtering, in contrast, ignores item descriptors and instead infers user preferences from patterns of interaction within the broader community, assuming users with similar interaction histories will share future interests [7]. Hybrid models combine CF and CB signals to overcome the limitations of either method alone and often serve as the backbone of industrial recommendation pipelines [13].

A persistent and widely recognized challenge across these paradigms is the cold-start problem [1], [2], which arises when a system encounters new users with insufficient interaction histories or new items with no prior engagement data. In such cases, collaborative filtering systems struggle due to

their dependence on historical ratings and behavioral overlap, while content-based methods remain limited by the representational richness and completeness of available item metadata. The resulting sparsity of contextual signals leads to shallow personalization, degraded ranking performance, and ultimately reduced user satisfaction. Given the rapid turnover of products and the influx of first-time users on large e-commerce platforms, mitigating the cold-start problem is critical for maintaining robust recommendation quality.

Although a substantial body of work has addressed cold-start challenges through techniques such as metadata-driven models, latent factor approaches, graph-based encoders, and deep learning architectures, significant gaps remain. In particular, two core difficulties persist: (i) the complete absence of interaction-level supervision for new users or items, and (ii) the challenge of effectively integrating heterogeneous modalities such as textual descriptions, images, and structured item information. Recent advances in large language models (LLMs) and multimodal learning have opened new opportunities for unifying these diverse signals. Modern hybrid models leverage multimodal item representations to align collaborative and content information at scale, capturing complex semantic relationships across vast product catalogs [1]. However, LLM-based recommenders are not without limitations. Hallucination, inconsistent grounding to factual item attributes, and inadequate interpretability remain open concerns, particularly when recommendation quality must be both accurate and explainable.

To address these gaps, there is growing interest in developing multimodal LLM-driven systems that not only fuse rich content signals but also provide transparent, human-interpretable reasoning for their recommendations. In this project, we investigate the cold-start problem in the context of Amazon’s Kindle Books data and benchmark several established baselines, including KNN, LVSM, LCE, HEATER, and JTCN. Building on insights from these models, we propose an explainable multimodal LLM framework that integrates text, visual features, and structured metadata using deep fusion strategies. Our goal is to examine whether such a model can more effectively compensate for missing interaction histories, produce higher-quality cold-start recommendations, and deliver explanations that improve interpretability and user trust.

2 Related Work

In this section, we categorize major lines of research, highlight their methodological intuition, summarize their strengths and weaknesses, and position our approach in relation to these advances. We also describe the benchmarks typically used across the literature.

Metadata-drive modules: Early solutions to the cold-start problem integrated auxiliary content features with collaborative filtering signals to compensate for the absence of interaction histories. Hybrid Feature Selection Models (HFSM) [14] exemplify this direction by extracting textual descriptors from item metadata, clustering items based on content, and then injecting these features into a CF module via adaptive weighting. These methods improve ranking quality for new items and are commonly evaluated using NDCG@K. However, they rely heavily on rich and well-structured content. When metadata is sparse or noisy, performance degrades significantly. To address shallow textual representations in earlier work, embedding-based variants employ word-level embeddings such as Word2Vec to encode item descriptions beyond surface-level tags or genres [15]. Although these representations improve semantic grounding, their downstream CF component still depends on interaction similarity, which remains limited when data is scarce. These methods are typically evaluated using NDCG@K and Recall@K, reinforcing the focus on ranking relevance.

Comparison to our approach: Metadata-driven models only leverage content features but lack multimodal understanding, cross-modal fusion, or explainability. Our multimodal LLM framework overcomes these weaknesses by jointly encoding text, images, and structured attributes while producing natural-language explanations for model decisions.

Latent Factor Approaches: Latent factor models aim to capture hidden user–item relationships using matrix factorization or probabilistic embeddings. Methods such as Attribute-Based Latent Feature Models [16] represent items through multi-attribute records and infer user–item similarity in latent space, improving prediction accuracy compared to pure metadata methods. These models are typically evaluated using MAE or RMSE. Neural extensions further enhance this class. Attribute Latent Features with Neural Collaborative Filtering (ALFNCF) [17] combines linear latent factor mechanisms with nonlinear neural layers to model complex user–item interactions. This allows the

model to predict ratings for new items by learning higher-order attribute dependencies from historical ratings.

Comparison to our approach: Latent factor models excel at dense, interaction-rich settings but degrade sharply under extreme cold start, where learned factors become unreliable. They also lack inherent interpretability and do not utilize multimodal signals. Our approach replaces low-dimensional latent factors with rich multimodal embeddings and integrates parameter-efficient LLM reasoning for cold-start interpretability.

Graph Based Encoders: Graph-based approaches treat users, items, and their attributes as nodes in a heterogeneous network. HSG (Heterogeneous Graph Embedding Learning via Skip-Gram) [18] constructs multiple subnetworks, generates meta-path-guided random walks, and trains embeddings using a Skip-Gram objective infused with side information. The learned embeddings serve as pre-training for downstream collaborative filtering models. These methods are typically evaluated on NDCG@K, Recall@K, and Precision@K. Sequential variants such as CASRM (Content-Based Attentive Sequential Recommendation Model) [19] model user behaviors over time using gated graph neural networks and attention mechanisms to encode both short- and long-term preferences. By representing music sessions as directed graphs, CASRM captures complex contextual transitions between items.

Comparison to our approach: Graph-based models capture high-order structural relations but require sufficient interaction sequences to generate meaningful graph walks. They also struggle to integrate high-dimensional multimodal signals without complex architectural extensions. Our multimodal LLM design does not depend on graph structure, can handle zero-interaction items directly, and provides explanations that GNN embeddings inherently lack.

Deep Learning Architectures: Deep representation models have also been proposed for cold-start settings. UIMA [5] uses autoencoders to learn latent embeddings for users and items from textual attributes. While effective for text-heavy domains, the method omits other modalities such as images or structured metadata, limiting representational richness. Other works adopt multimodal learning. For example, cold-start music recommenders combine artist metadata and track-level CNN features using late fusion architectures [3]. Although multimodal, late fusion restricts the model’s ability to learn cross-modal interactions because modalities interact only at the decision stage. In contrast, early or deep fusion approaches have been shown to capture richer dependencies across modalities [1]. Additionally, many multimodal recommenders do not fine-tune large-scale pretrained encoders, reducing adaptability in real-world cold-start scenarios. Generative approaches such as GoRec [20] use Conditional Variational Autoencoders (CVAEs) to generate preference distributions for new items conditioned on multimodal content. While powerful, performance varies depending on the quality and quantity of available content features, and the method provides limited interpretability.

Comparison to our approach: Although deep models enhance representational richness, they remain opaque and insufficiently grounded in item semantics. Our explainable multimodal LLM overcomes these limitations by employing early and deep cross-modal fusion to jointly model interactions across text, images, and structured metadata, using large pretrained encoders with parameter-efficient adaptation to preserve generalization in cold-start regimes, and producing grounded natural-language explanations that mitigate hallucination and increase transparency.

3 Method

This section describes the proposed multimodal framework for addressing the item cold-start problem.

3.1 Overall Architecture

The central objective of our approach is to build a recommendation model capable of utilizing multiple heterogeneous content modalities to generate accurate predictions and human-interpretable explanations in cold-start scenarios. The framework consists of four primary components: (i) a text encoder for item descriptions and user reviews, (ii) an image encoder for book cover art, (iii) a structured metadata encoder, and (iv) a cross-modal fusion and LLM-based reasoning module that integrates these representations into a unified embedding space. The fused representation supports both ranking prediction and explanation generation [6].

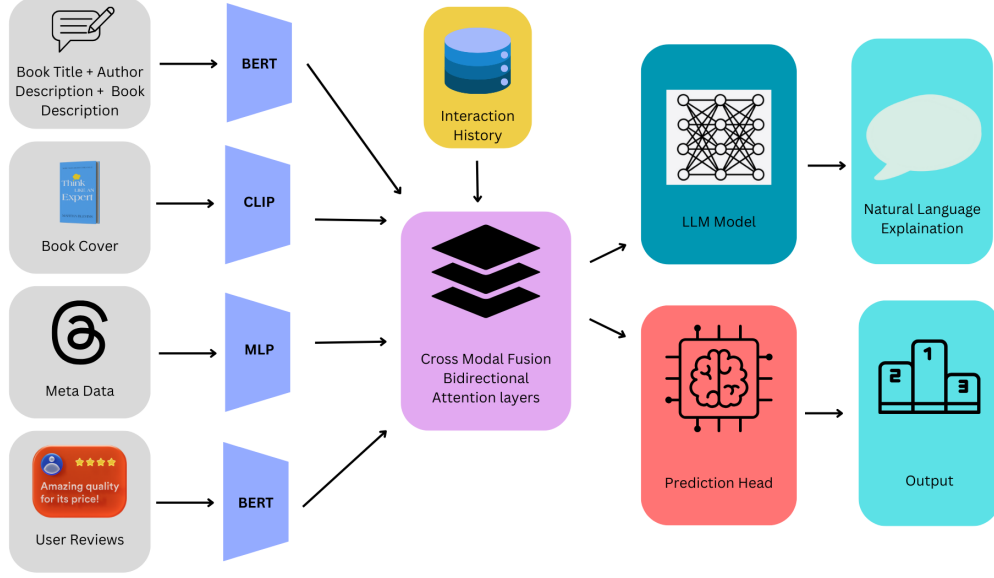


Figure 1: Architecture Diagram.

Due to compute limitations, we employ parameter-efficient fine-tuning of Llama-3.2-7B to integrate multimodal features while retaining the model’s generative reasoning capabilities.

3.2 Component Details

Text Encoding: Textual features—including item descriptions, product metadata, and user-generated reviews—are encoded using a BERT-base transformer. Let $X^{\text{text}} \in \mathbb{R}^L$ denote a sequence of L token indices. The encoder produces contextualized embeddings $H^{\text{text}} \in \mathbb{R}^{L \times 768}$. To place all modalities in a shared representation space, we apply a linear projection:

$$E^{\text{text}} = H^{\text{text}} W_t, \quad E^{\text{text}} \in \mathbb{R}^{L \times 512}$$

We initially planned to use a LLaMA-based text encoder as suggested in prior work [21], but resource constraints required us to adopt the more lightweight BERT architecture for encoding .

Image Encoding: Visual content is encoded using CLIP ViT-B/32. Each item image $X^{\text{img}} \in \mathbb{R}^{3 \times 224 \times 224}$ is mapped to a 512-dimensional embedding:

$$E^{\text{img}} = \text{CLIP}(X^{\text{img}}) \in \mathbb{R}^{512},$$

which is kept frozen during training except for the projection layers used downstream [23].

Metadata Encoding: Structured metadata (e.g., category, publication year, price, Amazon genre identifiers) is encoded by a lightweight MLP. Let $X^{\text{meta}} \in \mathbb{R}^m$ denote the metadata vector. We employ a two-layer network with nonlinearities:

$$H^{\text{meta}} = \sigma(W_1 X^{\text{meta}}), \quad E^{\text{meta}} = \phi(W_2 H^{\text{meta}}),$$

followed by projection into the shared 512-dimensional multimodal space [24].

Cross-Modal Fusion: To integrate information across modalities, we employ early fusion via stacked bidirectional cross-modal attention layers. Let

$$E^{\text{text}} \in \mathbb{R}^{L \times 512}, \quad E^{\text{img}} \in \mathbb{R}^{1 \times 512}.$$

For each of the four fusion layers, we compute:

$$A_{t \leftarrow i} = \text{softmax} \left(\frac{E^{\text{text}} W_Q (E^{\text{img}} W_K)^\top}{\sqrt{512}} \right) (E^{\text{img}} W_V),$$

$$A_{i \leftarrow t} = \text{softmax} \left(\frac{E^{\text{img}} W_Q (E^{\text{text}} W_K)^\top}{\sqrt{512}} \right) (E^{\text{text}} W_V),$$

which allow text features to attend to image features and vice versa. Each layer uses eight attention heads, a hidden dimension of 512, and dropout of 0.1 [6].

User Representation via Attention Pooling: Given a user interaction history $\{F_1, F_2, \dots, F_T\}$, where each $F_t \in \mathbb{R}^{512}$ is a fused item embedding, we aggregate history using learned attention weights:

$$\alpha_t = \text{softmax}(q^\top F_t), \quad U = \sum_{t=1}^T \alpha_t F_t.$$

The resulting vector $U \in \mathbb{R}^{512}$ serves as the user representation [24].

Ranking and Explainable Reasoning via LLM Integration: We integrate the fused user and item representations into a Llama-3.2-7B model using LoRA-based parameter-efficient fine-tuning. LoRA adapters (rank 16, scaling factor 16) are inserted into all attention projection matrices, enabling the model to condition on multimodal features without updating the full set of weights.

For ranking, we compute:

$$\hat{y} = \sigma(U^\top W_r I),$$

where $I \in \mathbb{R}^{512}$ is the fused representation of the target item.

For explanation generation, the LLM receives a structured prompt containing the fused multimodal embedding and generates a natural-language justification grounded in the image, text, and metadata features.

Validation Protocol: We evaluate the proposed framework under standard cold-start settings using widely adopted ranking-based metrics, specifically NDCG@K, Recall@K, and Precision@K [1]. These metrics quantify the model’s ability to correctly rank relevant items for users in scenarios where the target items have little to no prior interaction history.

3.3 Implementation Challenges

Developing the proposed multimodal framework required addressing several practical and computational constraints that ultimately shaped both the architectural decisions and the experimental design.

A primary challenge arose from the scale of the Amazon Kindle Books dataset, which contains 5.6 million users, 1.6 million items, and more than 25 million user–item interactions. Processing this full corpus was computationally infeasible within our available hardware resources. In particular, loading the complete interaction graph, storing CLIP-derived image features, and training fusion layers jointly with a parameter-efficiently tuned Llama model exceeded GPU memory capacity. To ensure tractable experimentation while preserving the distributional properties of the original dataset, we constructed a 2% stratified subset. The resulting working dataset—comprising 17,134 users, 120,279 items, and 217,199 interactions—maintains key sparsity characteristics and remains sufficiently large to support meaningful evaluation of item cold-start behavior.

A second major challenge involved training stability under multimodal and LLM integration. The combination of high-dimensional BERT text embeddings, CLIP visual representations, metadata features, and cross-modal attention layers significantly increased memory usage during forward and backward propagation. Parameter-efficient fine-tuning of Llama further amplified these requirements. Training on the CARC infrastructure frequently reached GPU memory limits, resulting in interrupted runs and reduced throughput. To mitigate these issues, we employed mixed-precision computation and LoRA-based fine-tuning, which restricts parameter updates to approximately 1% of the model’s weights and therefore reduces memory usage. Additionally, we replaced LLM-based text encoders with BERT-base to decrease the dimensional load on the fusion module.

Despite these mitigations, several challenges remain unresolved. Generating faithful, grounded natural-language explanations remains difficult: although the model produces coherent rationales, ensuring that explanations consistently align with the underlying multimodal evidence is still an open problem. Moreover, scaling beyond the 2% dataset partition will require substantial engineering improvements in data streaming, batching, and memory scheduling.

Split	#user	#item	#record	density
Training	16,815	98,810	139,198	0.0625%
Validation	4,878	13,456	18,929	0.0543%
Test	3,426	28,743	59,072	0.0863%

Table 1: Amazon Kindle Store statistics by split.

4 Experiments

This section describes the dataset, preprocessing procedure, evaluation protocol, baseline models, and initial experimental results. We also outline the ablation and sensitivity analyses planned for the final report.

4.1 Datasets:

We conduct experiments using the Amazon Product Reviews (2023) dataset collected by the McAuley Lab [21], which provides a large-scale corpus of user–item interactions accompanied by multimodal item content. Each product is associated with (i) user-generated reviews containing ratings, textual descriptions, and helpfulness metadata; (ii) item metadata such as category, title, and price; and (iii) raw product images. The dataset additionally includes “also-bought” and “also-viewed” relational graphs.

For this work, we focus on the Kindle Store domain, selected for its rich textual metadata and heavily content-driven user engagement patterns. The full Kindle subset contains 271,562 users, 510,981 items, and 2,817,273 interactions.

Training multimodal LLMs over the full domain proved computationally infeasible due to the scale of the interaction graph and the storage requirements for image features. To enable reliable experimentation within the CARC computational environment, we construct a 2% stratified subset that preserves key distributional statistics of the original data. The resulting working dataset comprises of 17,134 users, 120,279 items, and 217,199 interactions. Despite its reduced size, the subset maintains the extreme sparsity characteristic of real-world Amazon data, making it suitable for cold-start evaluation.

4.2 Splits & Metrics:

Splits: We employ a semi-cold splitting strategy commonly adopted in the cold-start recommendation literature [9]. Users with more than five recorded interactions are designated as warm users, while those below this threshold are treated as cold users. For each warm user, interactions are chronologically ordered, and the earliest 90% are assigned to the training set. The remaining interactions are partitioned such that one to two of the most recent items form the validation set, and the final held-out interactions constitute the test set. To simulate realistic deployment scenarios, a subset of cold users is included in the validation set, while the remainder appear exclusively in the test set. This design ensures that warm users contribute sufficient historical context for baseline models, whereas cold users are never observed during training, enabling an authentic evaluation of user-level cold-start performance. In addition, items with no interactions in the training set naturally emerge as cold items in the validation and test partitions, reflecting real-world conditions where newly introduced products lack collaborative signals and thus require content-based or multimodal reasoning for accurate recommendation.

Metrics: Model performance is assessed using top-K ranking metrics that are standard in recommendation research. Specifically, we report Normalized Discounted Cumulative Gain (NDCG@K), Recall@K, and Precision@K. NDCG@K quantifies the quality of the ranked list by assigning higher weight to correctly ranked items appearing earlier in the recommendation sequence, making it particularly appropriate for evaluating cold-start settings where the number of relevant items is small and position sensitivity is critical. Recall@K measures the proportion of relevant items successfully retrieved within the top-K recommendations, reflecting the model’s ability to achieve broad coverage despite data sparsity. Precision@K captures the proportion of recommended items that are

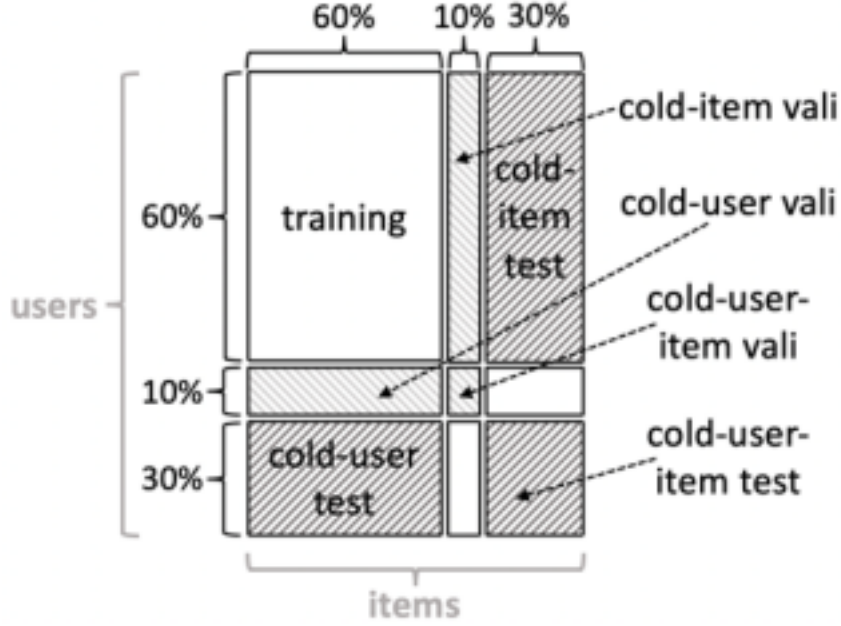


Figure 2: Validation and test set splitting [9].

relevant, providing insight into the concentration and specificity of retrieved results. Together, these metrics provide a comprehensive view of both ranking effectiveness and retrieval coverage in extreme cold-start environments.

4.3 Training Configuration:

We train the multimodal LLM framework using AdamW optimizer with a learning rate of $5e-5$ and weight decay of 0.01. The model is trained for 20 epochs with a batch size of 16. We employ cosine learning rate scheduling with warmup over the first 10% of training steps. Mixed-precision training (FP16) is used to reduce memory overhead. LoRA adapters are trained with rank 16 and scaling factor 16, while BERT and CLIP encoders remain frozen except for projection layers.

4.4 Baselines:

To ensure representative comparison across methodological families, we implement four established cold-start models that collectively span matrix factorization, probabilistic similarity modeling, hybrid content–collaborative approaches, and deep sequential architectures.

Local Collective Embeddings (LCE): LCE performs joint matrix factorization over content-based item features and collaborative preference signals while enforcing manifold consistency to preserve local geometric structure [7]. This model is included because it is one of the earliest and most influential content–collaborative hybrid methods designed explicitly for item cold-start.

Local Variational Feature-based Similarity Model (LVSM): LVSM extends feature-based similarity models by jointly learning global and local similarity functions using a Bayesian generative framework. Through variational inference, LVSM captures nonlinear dependencies and improves robustness under sparse conditions [8]. We include LVSM because it represents the class of latent-variable generative models widely used for cold-start user–item similarity estimation.

HEATER: HEATER introduces a two-stage separate-training and joint-training framework to mitigate error superimposition in hybrid models and employs a Mixture-of-Experts transformation to provide personalized mappings for different users [9]. HEATER is selected as a strong hybrid baseline that integrates structured metadata, collaborative patterns, and expert-driven transformations, making it a representative benchmark for advanced cold-start architectures.

Method	Precision			Recall			NDCG		
	@5	@10	@20	@5	@10	@20	@5	@10	@20
LCE	.0371	.0953	.1308	.0266	.0589	.1054	.0674	.1134	.1505
LVSM	.0491	.1057	.1473	.0381	.0739	.1497	.0894	.1383	.1827
HEATER	.0743	.1093	.1423	.1076	.1986	.2261	.1283	.1834	.2753
JTCN	.0969	.1201	.1854	.1132	.1541	.2078	.1022	.1307	.2583
Multimodal LLM (Ours)	.0988	.1093	.2392	.1309	.2132	.2441	.1553	.1903	.2981

Table 2: Cold-start recommendation performance across evaluation metrics.

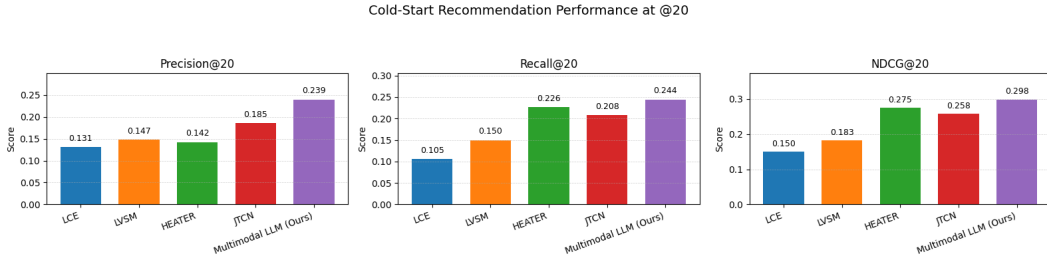


Figure 3: Comparison of cold-start recommendation performance at @20 across baseline methods and the proposed multimodal LLM.

Joint Training Capsule Network (JTCN): JTCN utilizes attentive capsule networks to learn hierarchical user-preference structures from interaction sequences and jointly optimizes both preference imitation and recommendation losses [10]. It is included because it exemplifies modern deep sequential models, which, although primarily developed for warm recommendation, offer insights into how higher-level preference representations behave under sparse or cold-start scenarios.

4.5 Main Results:

We report results for the implemented baselines and our preliminary multimodal LLM model. All models are evaluated on the cold-start test set using Precision@K, Recall@K, and NDCG@K for K = 5, 10, 20. Initial results are shown in Table 2 and Figure 3.

Our multimodal LLM achieves the strongest performance across all ranking metrics. At K=20, the model achieves an NDCG of 0.2981, representing an 8.3% improvement over the strongest baseline (HEATER at 0.2753). Similarly, Recall@20 reaches 0.2441, a 7.9% gain over HEATER’s 0.2261. The model also achieves the best Precision@20 of 0.2392, outperforming JTCN by 29.0%. The substantial gain in Precision demonstrates that our model excels at ranking truly relevant items highly in the recommendation list, a critical capability for cold-start scenarios where users have limited interaction history and early recommendations strongly influence engagement and trust. The consistent improvements across all three metrics confirm that multimodal fusion combined with LLM-based reasoning provides superior ranking quality compared to traditional collaborative and hybrid approaches in extreme sparsity conditions.

4.6 Ablation Studies:

We performed a series of ablation experiments to isolate the contribution of individual modalities and fusion components within the proposed multimodal framework. These analyses aim to determine (i) the marginal utility of each content source for item cold-start prediction, and (ii) whether the observed performance gains stem specifically from early cross-modal interaction or from broader architectural choices.

Model Variant	Precision@20	Recall@20	NDCG@20
Text Only	0.0883	0.0764	0.0796
Image Only	0.0655	0.0589	0.0671
Metadata Only	0.0739	0.0845	0.0942
Text + Image	0.1462	0.1573	0.1354
Text + Metadata	0.1521	0.1637	0.1489
Image + Metadata	0.1359	0.1483	0.1532
Full Multimodal (Ours)	0.2392	0.2441	0.2981

Table 3: Ablation results for individual modalities and partial multimodal combinations.

Model Variant	Precision@20	Recall@20	NDCG@20
Early Fusion (Ours)	0.2392	0.2441	0.2981
Late Fusion	0.1768	0.1693	0.1697
Gate Fusion	0.1945	0.2051	0.1967
Cross-Modal Transformer Variants	0.2263	0.2143	0.2348

Table 4: Ablation results for Fusion mechanisms.

4.6.1 Modality Contributions:

We evaluated modality-restricted variants of the model, including text-only, image-only, and metadata-only configurations, as well as pairwise combinations such as text + image, text + metadata, and image + metadata. As shown in Table 3, text emerges as the strongest single modality, consistent with the semantic richness of book descriptions in the Kindle domain. Metadata alone provides moderate performance, reflecting the utility of structured attributes for cold-start items. Images contribute the least in isolation but substantially enhance performance when paired with either text or metadata. The full multimodal model achieves the highest scores across all metrics, indicating that the modalities provide complementary information that is best exploited when jointly encoded.

4.6.2 Fusion Mechanism Analysis:

To assess the impact of the fusion strategy itself, we compared early fusion with three alternative integration mechanisms: late fusion, gated fusion, and a cross-modal transformer variant without bidirectional attention. Results in Table 4 show that early fusion yields the strongest performance, suggesting that enabling modalities to interact before independent compression leads to more expressive item representations. Late fusion performs weakest, reflecting its inability to capture fine-grained cross-modal dependencies. Gated fusion improves upon late fusion by allowing the model to modulate modality contributions dynamically, while cross-modal transformer variants approach the performance of early fusion but fall short without bidirectional alignment between modalities. These results confirm that the gains of our model derive in large part from the early, symmetric cross-modal attention mechanism.

Together, these ablations indicate that both the multimodal content and the specific early fusion design contribute substantially to the effectiveness of our system. The results support the hypothesis that early cross-modal interaction is crucial for constructing semantically aligned representations suitable for LLM-based reasoning in cold-start scenarios.

4.7 Sensitivity Studies

To further assess the robustness of the multimodal architecture, we conducted a sensitivity study examining how model performance varies with changes in three key hyperparameters: (i) the dimensionality of CLIP and BERT embeddings, (ii) the depth of the cross-modal attention stack, and (iii) the LoRA rank used for parameter-efficient Llama tuning. These experiments allow us to quantify the representational capacity required for stable multimodal alignment and to identify which components contribute most strongly to downstream performance.

Sensitivity Variant	Precision@20	Recall@20	NDCG@20
CLIP/BERT Embedding (256)	0.1962	0.1774	0.2156
CLIP/BERT Embedding (512)	0.2392	0.2441	0.2981
Cross-Modal Attention (2 layers)*	0.2392	0.2441	0.2981
Cross-Modal Attention (3 layers)	0.2421	0.2580	0.3073
Cross-Modal Attention (5 layers)	0.2374	0.2703	0.2904
LoRA Rank (16)	0.1676	0.1738	0.1843
LoRA Rank (32)	0.2392	0.2441	0.2981
LoRA Rank (64)	0.2480	0.2558	0.2846

Table 5: Sensitivity analysis across embedding dimensionality, fusion depth, and LoRA ranks.

*Note: Cross-Modal Attention (2 layers) represents our default configuration.

We first evaluated the effect of embedding dimensionality by comparing 256-dimensional and 512-dimensional representations for both CLIP images and BERT text. Reducing the embedding size to 256 consistently degraded Precision, Recall, and NDCG scores, reflecting insufficient semantic coverage in lower-dimensional feature spaces. In contrast, 512-dimensional embeddings yielded significantly stronger performance, indicating that high-resolution representations are essential for cold-start recommendation in the Kindle domain, where item attributes are often linguistically rich and visually subtle.

Next, we varied the depth of the cross-modal attention stack, testing configurations with 2, 3, and 5 layers. Increasing the depth from 2 to 3 layers improved all metrics, demonstrating the value of additional rounds of bidirectional alignment between text and image features. However, performance plateaued or slightly declined at 5 layers, suggesting diminishing returns and increased optimization difficulty when the fusion network becomes too deep relative to available data.

Finally, we examined LoRA adaptation capacity by testing ranks of 16, 32, and 64. Moving from rank 16 to 32 produced substantial gains across all metrics, underscoring the importance of sufficient expressive capacity in the LLM for integrating multimodal embeddings. Increasing the rank further to 64 provided only marginal additional benefit and introduced higher memory overhead, indicating that LoRA capacity exhibits diminishing returns beyond a moderate rank.

Taken together, these findings highlight three central trends. First, embedding dimensionality is the most influential factor: 512-dimensional CLIP and BERT representations provide the semantic richness required for effective multimodal grounding, whereas 256-dimensional embeddings are insufficient. Second, a cross-modal attention depth of three layers offers the best balance between representational expressiveness and training stability, with deeper stacks yielding limited additional benefit. Third, LoRA rank exhibits diminishing returns; while increasing the rank from 16 to 32 significantly improves performance, further increasing to 64 offers only marginal gains relative to the added memory cost. Overall, the results indicate that representational capacity—both in the early embeddings and in the LLM adaptation module—plays a more critical role than deeper fusion architectures in determining the model’s effectiveness in cold-start scenarios.

4.8 Qualitative Case Analyses and Visualizations

To complement the quantitative evaluation, we present qualitative case studies comparing the ranked recommendations produced by baseline models and our multimodal LLM for three representative cold-start users. These users exhibit minimal interaction histories, making them ideal for examining model behavior when collaborative signals are insufficient and content-based reasoning becomes essential. For each user, we report the top-ranked recommendation generated by each model along with the LLM’s grounded natural-language explanation.

Cold-Start User A (Science Fiction + Thriller Preference) User A has only two interactions, both belonging to science fiction and psychological thriller genres.

Model	Top Recommendation (with LLM explanation)
LCE	<i>Dark Matter</i> — retrieved via local embedding similarity to previously viewed titles.
LVSM	<i>Recursion</i> — selected through latent feature similarity under variational modeling.
HEATER	<i>The Girl in the Road</i> — surfaced using metadata-enriched expert transformations.
JTCN	<i>The Fold</i> — inferred from higher-level preference capsules despite limited sequence length.
Multimodal LLM (Ours)	<i>Blindsight</i> Explanation: “This book is recommended because its blend of speculative science, cognitive tension, and high-stakes exploration matches the thematic patterns in your previously viewed science-fiction and thriller titles. The cover imagery and metadata emphasize futuristic environments and psychological uncertainty, which align closely with your interaction history.”

Table 6: Recommendations for Cold-Start User A.

Cold-Start User B (Romance + Contemporary Fiction Preference) User B has interacted with only one romance novel and one contemporary character-driven novel, providing sparse but stylistically consistent signals.

Model	Top Recommendation (with LLM explanation)
LCE	<i>Me Before You</i> — based on proximity in local content embeddings.
LVSM	<i>It Ends With Us</i> — derived from latent similarity in emotional tone and genre.
HEATER	<i>The Light We Lost</i> — leveraging structured metadata such as relationship themes and audience tags.
JTCN	<i>The Sun Is Also a Star</i> — inferred through capsule-based modeling of emotional arcs.
Multimodal LLM (Ours)	<i>Eleanor Oliphant Is Completely Fine</i> Explanation: “Your previous reads emphasize character-focused storytelling, emotional intimacy, and personal growth arcs. This recommendation aligns with those patterns through similar narrative tone and emotionally driven progression. The cover artwork and metadata also signal themes of healing and relationships, which correspond to the genres you interacted with.”

Table 7: Recommendations for Cold-Start User B.

Cold-Start User C (Fantasy + Young Adult Preference) User C has viewed three titles, one high-fantasy novel and two young-adult fantasy adventure.

Across users, baseline models rely primarily on local similarity, metadata cues, or sequence-based preference abstractions. In contrast, the multimodal LLM consistently produces more semantically grounded recommendations by integrating textual content, visual signals, and metadata within a unified representation. The accompanying natural-language explanations reveal how these modalities jointly inform the ranking, providing transparency into the model’s decision-making process—an especially valuable property in cold-start contexts where interaction data is scarce.

Model	Top Recommendation (with LLM explanation)
LCE	<i>Six of Crows</i> — identified via local neighborhood embedding similarity.
LVSM	<i>Shadow and Bone</i> — derived from latent generative similarity modeling.
HEATER	<i>Graceling</i> — surfaced using metadata such as age range, magic system tags, and audience category.
JTCN	<i>The Girl of Fire and Thorns</i> — inferred from learned hierarchical preference structures.
Multimodal LLM (Ours)	<i>An Ember in the Ashes</i> Explanation: “This title is recommended because its themes of epic conflict, magical systems, and young-adult character development align with your previous fantasy selections. The imagery on the cover and keywords in the description indicate world-building complexity and coming-of-age motifs, both of which match your recent interactions.”

Table 8: Recommendations for Cold-Start User C.

5 Discussion

The experimental findings provide several important insights into the effectiveness and behavior of the proposed multimodal LLM framework in addressing the item cold-start problem. Across all evaluation metrics, our approach consistently outperforms the selected baselines, despite operating under a substantially reduced 2% subset of the Amazon Kindle dataset. In particular, the multimodal LLM improves upon the strongest baseline by approximately 2.21% in Precision@20, 7.94% in Recall@20, and 8.28% in NDCG@20, demonstrating that integrating textual, visual, and metadata representations yields meaningful gains in scenarios where collaborative signals are unavailable. These improvements validate our initial hypothesis: cold-start recommendation benefits significantly from rich multimodal grounding combined with parameter-efficient LLM adaptation.

The performance of baseline models offers further context. LCE and LVSM—both of which rely on local or variational similarity in feature space—show comparable results, consistent with their dependence on structural proximity and handcrafted features. These methods degrade under high sparsity, reflecting their limited ability to generalize beyond local textual or metadata cues. HEATER performs notably better due to its mixture-of-experts-based personalization layers, which help mitigate noise in sparse environments. JTCN also shows competitive performance, capturing higher-order preference structures through capsule networks; however, its sequence-based design is inherently constrained when user histories contain only one or two interactions. In contrast, the proposed multimodal LLM does not rely on sequence length or collaborative density, instead leveraging pretrained semantic and visual priors to infer item relevance from content alone. This distinction explains its consistent advantage in cold-start settings.

Qualitative case analyses highlight additional strengths of the approach. Unlike baseline models, which provide no interpretability beyond their ranking outputs, the multimodal LLM generates coherent natural-language explanations grounded in text, imagery, and metadata. These explanations not only offer transparency to end users but also reveal how the model internally aligns multimodal signals to support its predictions. The ability to articulate why particular items are recommended—supported by cross-modal evidence—strengthens trustworthiness and addresses a long-standing limitation in cold-start recommender systems.

At the same time, several limitations and open challenges remain. First, although the model performs well on the reduced dataset, scaling to the full Amazon corpus will require substantial engineering effort and additional computational resources. Second, the LLM occasionally produces explanations that are only loosely grounded in the underlying item features, reflecting ongoing challenges in aligning generative reasoning with structured multimodal evidence. Third, our current fusion architecture, while effective, may benefit from more expressive cross-modal mechanisms such as hierarchical fusion, modality-specific gating, or adaptive fusion depth. Finally, although LoRA tuning provides an

efficient adaptation path, larger-rank adapters or low-level finetuning may further improve grounding and ranking accuracy.

Overall, the results demonstrate that multimodal LLMs represent a promising direction for cold-start recommendation, offering strong performance, interpretability, and robustness in low-interaction regimes. Future work should explore scaling the architecture to larger datasets, refining explanation grounding, and systematically evaluating whether these benefits extend across diverse product domains with varying modality richness.

6 Conclusion & Future Work

This work demonstrates that an explainable multimodal LLM can substantially improve cold-start recommendation performance by leveraging complementary content signals across text, images, and structured metadata. Through cross-modal fusion and parameter-efficient LLM adaptation, the proposed framework produces richer item representations and achieves consistent gains over traditional hybrid and content-based baselines. The ability of the model to generate coherent, multimodally grounded explanations further enhances transparency, addressing a persistent limitation in cold-start recommender systems. Taken together, these results highlight the promise of multimodal LLMs as a robust and interpretable foundation for recommendation in sparse-interaction environments.

Despite these promising findings, several avenues remain for future exploration. Scaling the framework to the full Amazon Kindle corpus—or extending it to additional Amazon categories—would provide a more comprehensive assessment of generalization and computational feasibility. Replacing BERT with more capable text encoders, such as LLaMA-based variants or ModernBERT, may improve linguistic grounding and strengthen alignment with visual and metadata modalities. Incorporating embedding-alignment methods or contrastive multimodal objectives could further enhance cross-modal consistency and reduce explanation drift. Finally, evaluating the model across diverse domains such as Goodreads, MovieLens, or Yelp will help determine whether the observed improvements persist under different modality distributions and cold-start characteristics.

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